EXPLORING CUSTOMER BEHAVIOR THROUGH DATABASE ANALYSIS

SQL / MS-SQL Server Project

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Technology: SQL / MS-SQL Server

1. Introduction

This project explores customer behavior through database analysis for an online food ordering platform. SQL is used to analyze customer visits, spending, favorite products, sales trends, product performance and Gold membership behavior.

The analysis is intended to generate useful customer and product insights that can support data-driven business decisions.

2. Problem Statement

The project focuses on analyzing customer behavior on an online food ordering platform. The main areas are customer visits, customer spending, favorite menu items and loyalty-program behavior.

3. Objectives

- Analyze sales revenue generated by products.
- Identify top products based on sales revenue.
- Analyze customer spending and visit frequency.
- Identify first orders and most purchased products.
- Analyze Gold membership transactions.
- Compare sales revenue across years.
- Generate actionable business insights.

4. Database and Tables

Database name: OnlineFoodDelivery

Table	Purpose	Important Columns
Sales	Sales transactions	userid, created_date, product_id
Product	Product details	product_id, product_name, price
User_name	Customer names	userid, Names
Users	Customer signup data	userid, signup_date
Goldusers_Signup	Gold membership data	user_id, gold_signup_date

5. Methodology

- Created the OnlineFoodDelivery database in MS-SQL Server.
- Created the required tables.

- Inserted the records available in the supplied project material.
- Used JOIN operations to combine related tables.
- Used GROUP BY and aggregate functions such as COUNT and SUM.
- Used CTEs and Window Functions for advanced analysis.
- Used YEAR, MIN and DATEDIFF for date-based analysis.
- Used TOP, RANK, ROW_NUMBER and LAG where required.

6. SQL Analysis – Q1 to Q20

- Q1. Total sales revenue per product.
- Q2. Top 3 products by sales revenue.
- Q3. Number of users who signed up and took Gold membership.
- Q4. Revenue generated from Gold users.
- Q5. Total revenue from Gold users.
- Q6. How long each user has been a Gold user.
- Q7. Most popular product among Gold users.
- Q8. Total sales revenue each year.
- Q9. Sales revenue change over years.
- Q10. Average Gold signup compared to just signup.
- Q11. Number of orders by Gold member users.
- Q12. Total amount spent by each customer.
- Q13. Customer visit frequency.
- Q14. First order by each customer.
- Q15. Most purchased item and count.
- Q16. Most popular item for each customer.
- Q17. First item purchased after becoming Gold.
- Q18. Item purchased before becoming Gold.
- Q19. Total orders and amount spent before becoming Gold.
- Q20. Rank all transactions for each member while Gold.

7. SQL Concepts Used

- SELECT
- JOIN
- GROUP BY
- COUNT and SUM
- CTE
- Subqueries
- RANK()

- ROW_NUMBER()
- LAG()
- YEAR()
- MIN()
- DATEDIFF()
- TOP

8. Key Insights

- Product revenue analysis helps compare product performance.
- Customer spending analysis helps identify higher-value customers.
- Visit frequency helps understand customer engagement.
- Popular-product analysis can support menu and promotion decisions.
- Year-wise revenue analysis helps identify sales trends.
- Gold membership analysis can be used to study loyalty-program engagement.

9. Actionable Recommendations

- Promote products with stronger sales performance.
- Identify and retain high-value customers.
- Use visit frequency for customer engagement strategies.
- Use popular-product information for menu and marketing planning.
- Evaluate Gold membership using complete Gold-user transaction data.
- Use year-wise sales trends for business planning.

10. Tools and Technologies

MS-SQL Server, SQL, relational database concepts, JOINs, aggregate functions, CTEs and Window Functions.

11. Conclusion

The project demonstrates how MS-SQL Server can be used to explore customer behavior from transaction, product and customer data. SQL analysis provides a structured way to understand sales, spending, visit patterns, popular products and loyalty-program behavior.

12. Project Limitation

The project material states that the case study contains 62 Sales records and 5 Gold members, but the complete readable Goldusers_Signup records are not available in the supplied material used during this work. Therefore, Gold-membership results that require those missing rows should not be presented as complete numerical findings.